

MULTIPLE NUISANCE BREAKS AND SPURIOUS PREDICTABILITY UNDER PERSISTENT REGRESSORS

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Supplementary Material

This supplement gives the technical details behind the block-sieve empirical likelihood construction in the main manuscript. Section S1 records the finite-sample block projection identities. Section S2 proves the stable-score decomposition and the omitted-break failure theorem. Section S3 proves the known-partition Wilks theorem by extending the original profile-sieve geometry to a regime-specific nuisance space. Section S4 gives the estimated-partition equivalence argument based on the set of misclassified observations. Section S5 proves the local-alternative calculation. Section S6 records implementation details for the reported simulations and the empirical application.

S1 Block Projection Algebra

Fix a partition $\Lambda = (k_1, \dots, k_q)$. Let $\mathbf{Q}_{K,\Lambda}$ and $\mathbf{M}_{K,\Lambda}$ be defined as in (3.1) and (3.2). Throughout this supplement, all projection equalities are

equalities in the empirical Hilbert space $(\mathbb{R}^T, T^{-1}\mathbf{a}'\mathbf{b})$.

Lemma S1 (Block projection identities). *If $\mathbf{Q}'_{K,\Lambda}\mathbf{Q}_{K,\Lambda}$ is nonsingular, then*

$$\mathbf{Q}'_{K,\Lambda}\mathbf{M}_{K,\Lambda} = \mathbf{0}, \quad \mathbf{M}'_{K,\Lambda} = \mathbf{M}_{K,\Lambda}, \quad \mathbf{M}_{K,\Lambda}^2 = \mathbf{M}_{K,\Lambda}.$$

For every $\mathbf{c}$ conformable with $\mathbf{Q}_{K,\Lambda}$,

$$(\mathbf{Q}_{K,\Lambda}\mathbf{c})'\mathbf{M}_{K,\Lambda}\mathbf{w} = 0.$$

Proof. Let $\mathbf{\Pi}_{K,\Lambda} = \mathbf{Q}_{K,\Lambda}(\mathbf{Q}'_{K,\Lambda}\mathbf{Q}_{K,\Lambda})^{-1}\mathbf{Q}'_{K,\Lambda}$. This is the orthogonal projector onto the column space of $\mathbf{Q}_{K,\Lambda}$. Therefore $\mathbf{\Pi}'_{K,\Lambda} = \mathbf{\Pi}_{K,\Lambda}$, $\mathbf{\Pi}_{K,\Lambda}^2 = \mathbf{\Pi}_{K,\Lambda}$, and $\mathbf{Q}'_{K,\Lambda}(\mathbf{I}_T - \mathbf{\Pi}_{K,\Lambda}) = \mathbf{0}$. Since $\mathbf{M}_{K,\Lambda} = \mathbf{I}_T - \mathbf{\Pi}_{K,\Lambda}$, the displayed identities follow. The final equality is obtained by multiplying $\mathbf{Q}'_{K,\Lambda}\mathbf{M}_{K,\Lambda} = \mathbf{0}$ by $\mathbf{c}$ and $\mathbf{w}$. $\square$

S2 Stable-Score Failure

This section proves Proposition 1 and Theorem 1. Let $\Lambda = 0$ denote the no-break partition and write $\mathbf{M}_{K,0}$ for the corresponding stable-nuisance residual-maker.

Proof of Proposition 1. Under $H_0 : \beta = \beta_0$,

$$\mathbf{Y} - \beta_0\mathbf{X}_{lag} = \boldsymbol{\mu} + \mathbf{U}.$$

Multiplying by the stable residualized weight gives the exact identity

$$T^{-1/2}(\mathbf{Y} - \beta_0 \mathbf{X}_{lag})' \mathbf{M}_{K,0} \mathbf{w} = T^{-1/2} \mathbf{U}' \mathbf{M}_{K,0} \mathbf{w} + T^{-1/2} \boldsymbol{\mu}' \mathbf{M}_{K,0} \mathbf{w}.$$

The second term is the omitted-break component. No stable-sieve approximation remainder is needed because B_T is defined using the exact nuisance vector $\boldsymbol{\mu}$. $\square$

Proof of Theorem 1. The scalar EL expansion gives

$$\ell_T^{\text{st}}(\beta_0) = \frac{\{S_T^{\text{st}}(\beta_0)\}^2}{V_T^{\text{st}}} + o_p(1).$$

By Proposition 1, $S_T^{\text{st}}(\beta_0) = A_T^{\text{st}} + B_T$. If $|B_T|/(V_T^{\text{st}})^{1/2} \rightarrow_p \infty$, the quadratic ratio diverges in probability, so rejection at any fixed critical value has probability tending to one. If the normalized oracle term converges to $N(0, 1)$ and the normalized break term converges to a nondegenerate random variable B , then the limiting quadratic form is $(Z + B)^2$, where $Z \sim N(0, 1)$ conditionally on the limiting variance. This is central chi-square only when $B = 0$ with probability one. $\square$

S3 Known-Partition Wilks Theorem

Set $\Lambda = \Lambda_0$ and write

$$\mathbf{v}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{w}, \quad \hat{\mathbf{u}}_0(\beta_0) = \mathbf{M}_{K, \Lambda_0} (\mathbf{Y} - \beta_0 \mathbf{X}_{lag}).$$

Under the true partition,

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \mathbf{Q}_{K, \Lambda_0} \mathbf{c}_0 + \mathbf{r}_0 + \mathbf{U},$$

where $\mathbf{r}_0$ stacks the regime-specific sieve approximation errors.

Lemma S2 (Known-partition replacement). *Under Assumptions 3 and 4,*

$$T^{-1/2} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^T U_t v_{t,0} + o_p(1), \quad (\text{S3.1})$$

$$T^{-1} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0)^2 = T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2 + o_p(1), \quad (\text{S3.2})$$

$$\max_{t \leq T} |\widehat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}). \quad (\text{S3.3})$$

Proof. Using Lemma S1,

$$\widehat{\mathbf{u}}_0(\beta_0) = \mathbf{M}_{K, \Lambda_0}(\mathbf{U} + \mathbf{r}_0), \quad \mathbf{v}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{w}.$$

Because $\mathbf{M}_{K, \Lambda_0}$ is symmetric and idempotent,

$$T^{-1/2} \widehat{\mathbf{u}}_0(\beta_0)' \mathbf{v}_0 = T^{-1/2} (\mathbf{U} + \mathbf{r}_0)' \mathbf{v}_0.$$

The approximation term is bounded by

$$|T^{-1/2} \mathbf{r}_0' \mathbf{v}_0| \leq \sqrt{T} \|\mathbf{r}_0\|_T \|\mathbf{v}_0\|_T = o_p(1),$$

which proves (S3.1). For the denominator, write $\widetilde{\mathbf{U}}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{U}$. The feasible score decomposes as

$$\widehat{Z}_t(\beta_0, \Lambda_0) = \widetilde{U}_{0,t} v_{t,0} + (\mathbf{M}_{K, \Lambda_0} \mathbf{r}_0)_t v_{t,0}.$$

The approximation contribution is negligible by Assumption 3. Assumption 4 then gives

$$T^{-1} \sum_{t=1}^T \{(\tilde{U}_{0,t} v_{t,0})^2 - (U_t v_{t,0})^2\} = o_p(1),$$

which proves the denominator equivalence (S3.2). The maximum bound follows from the oracle maximum condition, the projected-noise replacement condition, and the sup-norm sieve approximation bound. $\square$

Lemma S3 (EL expansion). *Let $Z_{t,T}$ be scalar scores. Suppose the convex-hull condition holds and*

$$T^{-1/2} \sum_t Z_{t,T} = O_p(1), \quad T^{-1} \sum_t Z_{t,T}^2 \rightarrow_p \eta^2, \quad \max_t |Z_{t,T}| = o_p(\sqrt{T}),$$

where $P(\eta^2 > \underline{\eta}) > 1 - o(1)$. Then

$$2 \sum_{t=1}^T \log(1 + \hat{\lambda} Z_{t,T}) = \frac{\{T^{-1/2} \sum_t Z_{t,T}\}^2}{T^{-1} \sum_t Z_{t,T}^2} + o_p(1).$$

Proof. The multiplier equation implies $\hat{\lambda} = O_p(T^{-1/2})$. Since $\max_t |\hat{\lambda} Z_{t,T}| = o_p(1)$, a Taylor expansion of the multiplier equation gives

$$\hat{\lambda} = \frac{\sum_t Z_{t,T}}{\sum_t Z_{t,T}^2} + o_p(T^{-1/2}).$$

Substitution into the second-order expansion of the log empirical likelihood gives the displayed quadratic approximation. $\square$

Proof of Theorem 2. Apply Lemma S2 to reduce the feasible score to the matched block-projection oracle score $U_t v_{t,0}$. Assumption 4 gives the mixed-

normal numerator limit, denominator consistency, and maximum-score condition. Lemma S3 then yields

$$\ell_T(\beta_0, \Lambda_0) = \frac{\{T^{-1/2} \sum_t U_t v_{t,0}\}^2}{T^{-1} \sum_t U_t^2 v_{t,0}^2} + o_p(1) \Rightarrow \chi_1^2.$$

□

S4 Estimated-Partition Equivalence

This section records the high-level argument behind Theorem 3. Let $\hat{\Lambda} = \hat{\Lambda}(\beta_0)$ and define

$$\mathcal{M}_T = \{t : j_{\hat{\Lambda}}(t) \neq j_{\Lambda_0}(t)\}.$$

Assumption 5 states that $|\mathcal{M}_T| = o_p(\sqrt{T})$ and that the boundary observations are negligible at the numerator and denominator scales.

Lemma S4 (Misclassification bound). *Suppose Assumption 5 holds. Suppose also that the projected weights are uniformly bounded in probability, the score arrays satisfy the same maximum-score bounds as in Assumption 4, and the segment Gram matrices are stable uniformly over admissible*

partitions. Then

$$\begin{aligned} T^{-1/2} \sum_{t=1}^T \{\widehat{Z}_t(\beta_0, \widehat{\Lambda}) - \widehat{Z}_t(\beta_0, \Lambda_0)\} &= o_p(1), \\ T^{-1} \sum_{t=1}^T \{\widehat{Z}_t(\beta_0, \widehat{\Lambda})^2 - \widehat{Z}_t(\beta_0, \Lambda_0)^2\} &= o_p(1), \\ \max_t |\widehat{Z}_t(\beta_0, \widehat{\Lambda})| &= o_p(\sqrt{T}). \end{aligned}$$

Proof. Away from $\mathcal{M}_T$, the estimated partition and true partition assign observations to the same regime. The two residual makers differ only through least-squares fits perturbed by the boundary observations and by terms controlled by uniform Gram stability. With $\mathcal{Z}_T$ defined in Assumption 5, the direct contribution of the boundary set to the numerator is bounded by

$$T^{-1/2} |\mathcal{M}_T| \mathcal{Z}_T = o_p(1).$$

The denominator contribution is bounded by $T^{-1} |\mathcal{M}_T| \mathcal{Z}_T^2 = o_p(1)$. The remaining terms are the uniform Gram-stability perturbations of the least-squares coefficients and are of the same smaller order. Uniform Gram stability transfers the maximum-score condition from the true partition to the estimated partition. $\square$

Proof of Theorem 3. Lemma S4 shows that the estimated-partition score and the true-partition score have the same first-order numerator, denominator, and maximum-score behavior. Applying Lemma S3 to both arrays

yields

$$\ell_T\{\beta_0, \widehat{\Lambda}(\beta_0)\} - \ell_T(\beta_0, \Lambda_0) = o_p(1).$$

The Wilks limit follows from Theorem 2. If $P\{\widehat{q}(\beta_0) = q_0\} \rightarrow 1$, the same argument holds on the event of correct order selection, proving Corollary 1. $\square$

S5 Local Predictive Alternatives

This section proves Theorem 4. Under the local alternative $\beta_T = \beta_0 + d_T$, the null-imposed residual satisfies

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \boldsymbol{\mu} + \mathbf{U} + d_T \mathbf{X}_{lag}.$$

At the true partition, the block projection removes the regime-specific sieve approximation to $\boldsymbol{\mu}$, so the score numerator has the expansion

$$T^{-1/2} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^T U_t v_{t,0} + d_T T^{-1/2} \mathbf{X}'_{lag} \mathbf{M}_{K, \Lambda_0} \mathbf{w} + o_p(1).$$

By the local drift condition in Theorem 4, the second term converges to $\eta_q \delta_h$.

The denominator remains $\eta_q^2 + o_p(1)$ because the local drift is at the testing scale and the nuisance-break component is removed by $\mathbf{M}_{K, \Lambda_0}$. Hence the quadratic EL expansion gives a $\chi_1^2(\delta_h^2)$ limit for the known partition.

Lemma S4 transfers the same numerator, denominator, and maximum-score behavior to $\widehat{\Lambda}(\beta_0)$, proving the estimated-partition version.

S6 Implementation Details

For simulations, each replication uses the same null-imposed profile path as the theory:

$$\beta \rightarrow \hat{q}(\beta), \hat{\Lambda}(\beta) \rightarrow \mathbf{M}_{K, \hat{\Lambda}(\beta)} \rightarrow \hat{Z}_t(\beta) \rightarrow \ell_T^{BA}(\beta).$$

The proof object is global profile-sieve segmentation. Computation may use multiscale candidate generation, followed by final selection with the profile-sieve RSS and the information criterion in (4.4). The implementation must report the candidate-generation method separately from the statistical estimator, because SBS, NOT, and related devices are computational accelerators rather than the main contribution.

For empirical work, the stable and break-adaptive statistics are reported together. The stable statistic documents conventional evidence of predictability; the adaptive statistic documents whether that evidence survives multiple nuisance breaks. The empirical section also reports $\hat{q}$, break dates, regime-specific nuisance fits, an estimated break-induced score component, and residualized predictor strength $T^{-1}\mathbf{w}'\mathbf{M}_{K, \hat{\Lambda}}\mathbf{w}$.